MILITARY SPECIFICATION

ELECTRONIC COMPONENT ASSEMBLY: 11737477 ("A" TRIGGER)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of electronic assembly known as Electronic Component Assembly: 11737477 ("A" Trigger) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-I-45607	Inspection Equipment: Acquisition, Maintenance, and Disposition of
MIL-S-45743	Soldering, Manual Type, High Reliability Electrical, Electronic, Instrument, Communication, and Radar for Aerospace and Control Systems, Procedure for

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MIL-S-46844	Solder Bath Soldering of Printed Wiring Assemblies, Automatic Machine Type

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

Fl1737477	Electronic Component Assembly - ("A" Trigger)

PACKAGING DATA SHEET

11737477	Packaging of Electronic Component Assembly - (“A” Trigger): 11737477

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fabrication. The Electronic Component Assembly ("A" Trigger), herein referred to as the assembly, shall be manufactured in accordance with Drawing F11737477 and drawings pertaining thereto.

3.2 General specification. The assembly shall meet the following requirements, where applicable, of MIL-F-13926:

a. Order of precedence

b. Dimensions and tolerances

c. Part identification and marking

d. Electrical and electronic assemblies

e. Workmanship.

3.2.1 Soldering. Soldering shall be in accordance with MIL-S-45743 or MIL-S-46844, except that bare copper wire shall be permitted at clipped ends of component leads.

3.3 Performance. Unless otherwise specified, the assembly shall meet the performance requirements of this specification at standard ambient temperatures between 60 degrees Fahrenheit (°F) and 90°F.

3.3.1 Loads, power, and signals. The assembly shall perform as specified herein when the loads, power and signals of table I are applied as specified.

3.3.1.1 Reticle light control. With the voltage signal source (item 3.1) of tabel adjusted to apply +14.0 ±0.5 Vdc between P1-3 and P1-6, the outputs at P1-4 and P1-9 shall each be +11.0 ±2.0 Vdc. With the item 3.1 signal source adjusted to apply +5.0 ±0.2 Vdc between P1-3 and P1-6, the outputs at P1-4 and P1-9 shall each be +3.6 ±0.8 Vdc.

3.3.1.2 A-trigger. With no input signal applied to P1-10, the output at P1-5 shall be +5.0 ±1.0 Vdc. With the current signal (item 3.2) of table I applied to P1-10, the output at P1-5 shall be +0.2 ±0.2 Vdc.

3.3.2 Environmental

3.3.2.1 Storage temperature. The assembly shall meet the requirements of 3.3.1 at ambient temperature after exposure to and thermal stabilization at -80°F and +160°F.

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TABLE I

LOADS, POWER, AND SIGNALS

Item	Condition	Characteristics	Connections
1.	Loads		Connected between the following pins:
1.1	Resistor	300 ohms ± 5 percent	P1-4 and P1-6
1.2	Resistor	300 ohms ± 5 percent	P1-9 and P1-6
2.	Power sources		
2.1	+5 Vdc	Tolerance: ±0.2 Vdc Ripple: ≤ 25 mV p-p	P1-7(+) and P1-6(-)
2.2	+15 Vdc	Tolerance: ±0.7 Vdc Ripple: ≤ 25 mV p-p	P1-2(+) and P1-6(-)
3.	Signal sources		Applied between the following pins when specified herein:
3.1	Zero to +15 Vdc maximum	Adjustable. Ripple: ≤ 25 mV p-p	P1-3(+) and P1-6(-)
3.2	90 μA	Tolerance: ±10 μA	P1-10(+) and P1-6(-)

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3.3.2.2 Operating temperatures, The assembly shall meet the requirements of 3.3.1 while exposed to and thermally stabilized at -40°F and +125°F, subsequent to whit-h the requirements of 3.3.1 shall be met at ambient temperature

3.3.2.3 Vibration. The assembly shall show no physical failure and shall meet the requirements of 3.3.1 after being exposed to the following vibratory conditions:

a. Amplitude:	1/64 inch, constant (1/32 inch double amplitude)
b. Vibratory motion:	Simple harmonic
c. Sweep time:	5 to 55 to 5 Hertz in a five-minute period
d. Vibration cycling:	Three linear or logarithmic sweeps applied to each of the three mutually perpendicular axes.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-l3926 form a part of the quality assurance provisions to this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval. The requirements for first article approval and the responsibility (Gcvernment or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three assemblies. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and Procedures as used in regular production. All parts and materials including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure conformance to all the requirements of this specification. A record of this inspection, including certificates of conformance for materials, shall be submitted with the sample.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of this specification. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

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4.3 Inspection provisions

4.3.1 Submission of product. Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with inspection or sampling procedures herein and submission of product provisions of MIL-STD-105.

4.3.2 Examination and tests

4.3.2.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the inspection provisions contained in the respective specifications and Supplementary (TDP). Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable quality assurance provisions of MIL-F-13426 shall apply.

4.3.2.2 Functional tests. The requirements and tests in table II shall be inspected on a 100 percent basis. All examinations and tests shall be conducted at the standard ambient temperature.

TABLE II

FUNCTIONAL TESTS

Classification	Characteristics	Requirements	Test Paragraphs
101.	Reticle light control	3.3.1.1	4.6.2
102.	A-trigger	3.3.1.2	4.6.3

4.4 Special sampling. The selection of samples shall be in accordance with section 7 of MIL-STD-105.

4.4.1 General. One assembly shall be selected at random by a Government representative as a control sample from each 50 produced. The control sample shall meet the requirements and tests in table III.

4.4.2 Environmental. Three assemblies shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests in table IV.

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TABLE III
SPECIAL SAMPLING - GENERAL

Classification	Control Tests	Requirements	Test Paragraphs
301.	Fabrication	3.1	Applicable drawings-visual
302.	General specification	3.2	MIL-F-13926-visual

TABLE IV
SPECIAL SAMPLING - ENVIRONMENTAL

Classification	Control Tests	Requirements	Test Paragraphs
303.	Storage temperature	3.3.2.1	4.6.4.1
304.	Operating temperature	3.3.2.2	4.6.4.1
305.	Vibration	3.3.2.3	4.6.4.2

4.4.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejection lots shall be in accordance with the acceptance and rejection criteria specified in MIL-STD-105.

4.5 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607. The inspection equipment listed in table V shall be utilized in performing the tests specified herein.

4.5.1 Government furnished inspection equipment. Where the contract provides for Government furnished inspection equipment, supply and maintenance of inspection equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

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TABLE V
INSPECTION EQUIPMENT

Item	Nomenclature	Description
1.	DC Power Supply	Zero to +15 Vdc maximum, adjustable Ripple: ≤ 100 mV p-p Source impedance: < 0.01 ohm
2.	DC Power Supply	+5.0 ±0.2 Vdc Ripple: ≤ 25 mV p-p Source impedance: < 0.1 ohm
3.	DC Power Supply	+15.0 ±0.7 Vdc Ripple: ≤ 25 mV p-p
4.	DC Power Supply	90 ±10 μA
5.	DC Voltmeter	0 to 50 Vdc Accuracy: ±1 percent

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAPs forming a p-art of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the TDP.

4.5.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the test methods and procedures specified in 4. 6 of this specification and with the component inspection procedures specified in examination and test facilities provisions of MIL-F-13926. Since tolerance of inspection equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the inspection equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of inspection equipment shall be so selected and controlled as to insure that the inspection equipment will reliably indicate acceptability of the product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of inspection equipment,

4.6 Test methods and procedures.

4.6.1 Test conditions. Unless otherwise specified, the test conditions shall be in accordance with the test facilities provisions of MIL-F-13926.

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4.6.1.1. Initial conditions. Connect the loads and power to the assembly as specified in table X.

4.6.2 Reticle light control test. Connect the signal source (item 3.1 of table I) between P1-3 and P1-6, and set the voltage, in turn, to +14.0 ±0.5 Vdc and +5.0 ±0.2 Vdc. Measure the outputs at P1-4 and P1-9 for each input condition to determine compliance with 3.3.1.1.

4.6.3 A-trigger test. With no input applied to P1-10, measure the output at P1-5 to determine compliance with 3.3.1.2. Then apply the current signal (item 3.2 of table I) to P1-10, and measure the output at P1-5 to determine compliance with 3.3.1.2.

4.6.4 Environmental tests

4.6.4.1 Temperature. The test apparatus shall be in accordance with the applicable test facilities provisions of MIL-F-13926 except that the assembly may be mounted on the inside surface of the test chamber. P'erform this test in accordance with Procedure I of MIL-F-13926, with conditions C (-40°F) and E (+125°F), and with the rate of temperature change during cycling not greater than 8°F per minute. At each of the specified operating temperatures, perform the tests of 4.6.2 and 4.6.3 to determine compliance with 3.3.2.1 and 3.3.2.2.

4.6.4.2 Vibration. Perform this test utilizing the apparatus specified in Method 514.1 of MIL-STD-810 and in accordance with Part 1 of Procedure I therein, except that the vibratory conditions of 3.3.2.3 shall apply. Following vibration exposure, examine the assembly for structural failures, loose hardware, wire breakage, and abrasions. Perform the tests of 4.6.2 and 4.6.3 to determine compliance with 3.3.2.3.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 11737477. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The assembly is a module of the Laser Receive-Transmitter RT-1021( )/VVG-1, intended for use in the Fire Control Rangefindler AN/VVG-1. The assembly:

a. Provides a reticle light control output

b. Generates the A-trigger signal

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification

b. Applicable packaging data sheet number (see 5.1)

c. Selection of applicable levels of preservation, packaging, and packing

d. Applicable stock number

e. Requirement for First Article Submission.

CUSTODIAN:

Army - MU

PREPARING ACTIVITY:

Army - MU

PROJECT NO. 1240-A316

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DOCUMENT IDENTIFIER AND TITLE

MIL-E-50748(MU) Electronic Component Assembly: 11737477 ("A" Trigger)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

[ ] YES [ ] NO *(If "Yes", in what way?)*

4. REMARKS

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